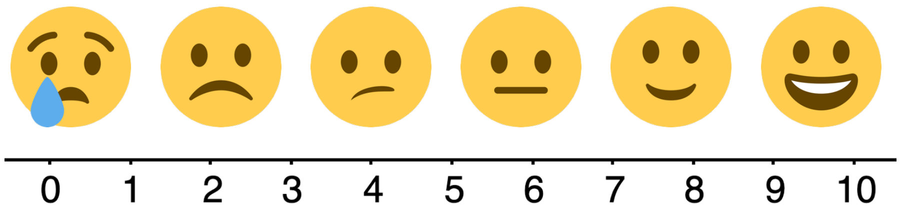
13.10.7 Wrap-Up: Reflection

1. Rate yourself using the scale below on today's learning target.

   I can verify using substitution that linear expressions are equivalent.



2. Write an explanation of why you rated yourself where you did.

Unit 13, Lesson 11: What Values Make an Equation or Inequality True?

Benchmark

MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false.

Learning Target

- ☐ I can determine if a given integer value makes an equation or inequality true.

13.11.1 Warm-Up: Day at Work – Film Composer

Shohan Cagle is a film composer. He writes music for commercials and soundtracks for TV shows and movies.

1. What did you notice about what he does to write a soundtrack?

2. Shohan said that most people say they don't normally hear the soundtrack. Name a TV show or a movie whose soundtrack you remember.

3. Shohan mentioned that one of his challenges is a short deadline. He said that his strategy to get his work done is "to sit and focus and get in the right mindset". How can you use Shohan's strategy?

13.11.2 Guided Instruction: Determining Values That Make an Equation True or False

An ***equation*** is a mathematical relation statement where two equivalent expressions and values are separated by an equal sign.

1. In the equation $4x + 5 = 8x - 23$, what are the two expressions?

2. Work with your partner to guess a value for x and then substitute it into both expressions. A high and a low guess have been provided for you in the table.

x	$4x + 5$	$8x - 23$	Are the expressions equal?
$^-3$	$4(^-3) + 5 = {}^-7$	$8(^-3) - 23 = -47$	No
14	$4(14) + 5 = 61$	$8(14) - 23 = 89$	No

For an equation to be **true**, both sides of the equation must have the same value. For example, the equation $78 - 56 = 2 \cdot 11$ is true because both expressions, $78 - 56$ and $2 \cdot 11$, are equal to 22.

The ***solution*** of an equation is the value(s) that, when substituted in for the variable(s), makes the equation true.

3. Based on your guesses, what is the solution to the equation $4x + 5 = 8x - 23$?

4. Use substitution to determine if the equation $2x + 4x + 7 = 19$ is true for each value.

x	$2x + 4x + 7 = 19$	Is the equation true?
-1		
0		
2		

5. The solution to the equation $2x + 4x + 7 = 19$ is $x =$ _____ because when this value is substituted for x, the equation is _________.

13.11.3 Guided Practice: Determining Which Values Make an Equation True or False

From the specified set of integers, determine which, if any, make the equation true. Show your work.

1. $18 + 7w = 10w - 9 \quad \{-9,\ -2,\ 4,\ 9,\ 11\}$

2. $-2(3 - p) = 4p + 10 \quad \{-8, -3, 0, 10\}$

13.11.4 Guided Instruction: Determining Values That Make an Inequality True or False

An ***inequality*** is a mathematical relationship between two expressions that are not equal ($<$, $>$, or $\neq$) or are not always equal ($\leq$ or $\geq$).

A ***strict inequality*** is a mathematical relationship between two expressions that are not equal. A strict inequality uses the symbols, $<$, $>$, or $\neq$.

1. Use substitution to evaluate the mathematical statement. Then determine if each inequality is true for the value of x.

x	Strict Inequality $9x - 5 < 13$	Inequality $9x - 5 \leq 13$	Which inequality is true?
5			
2			
$^{-}3$			

2. Answer the following.

   a. The inequality $x < 5$, means any number that is

      ○ smaller
      ○ larger

      than 5 will make the inequality true.

   b. List four numbers that are not integers that make the inequality true.

3. Answer the following.

   a. The inequality, $-11 \geq x$, means any number that is

      ○ smaller than or equal to
      ○ larger than or equal to

      $^{-}11$ will make the inequality true.

   b. List four numbers that are not integers that make the inequality true.

4. Use substitution to determine if the inequality $16 \geq 7m - 5$ is true for each value specified for m.

m	$16 \geq 7m - 5$	**Is the inequality true?**
-2	$16 \geq 7(-2) - 5$ $16 \geq -14 - 5$ $16 \geq -19$	Yes
0		
3		
6		

5. Answer the following.

   a. Determine the integer(s) from the specified set that make the inequality **true**. Show your work.

      $-x + 3 + 5x \geq {}^{-}8x - 7 \qquad \{-12, -5, 5, 16\}$

   b. When Diego substituted 16 in $-x + 3 + 5x \geq {}^{-}8x - 7$, he stopped after writing $-16 + 3 + 5(16) \geq {}^{-}8(16) - 7$ because he said he knew it was true. When his teacher asked him how he knew, he said it was because the left side would be a positive number and the right side would be a negative number.

      With your partner, discuss how Diego knew the left side of the inequality would be a positive number and the right side of the inequality would be a negative number.

6. Determine which values from the specified set make the inequality **false**. Show your work.

   $12 < 2(13y - 1) \qquad \{-14, 0, 1, 9\}$

   The values from the specified set that make the inequality $12 < 2(13y - 1)$, **false** are ____________________.

13.11.5 Your Turn!

1. Use substitution to determine if the inequality $86 \geq 17a - 8a - 23$ is true or false for each integer value specified for a.

a	$86 \geq 17a - 8a - 23$	True or False
$^{-}8$		
0		
7		
12		
21		

13.11.6 Wrap-Up: Reflection

1. One question I still have after today's lesson is …

____

____

____?

2. Today I am still unsure about …

____

____

____.

3. To fill in this gap I intend to …

____

____

____.